

in the same mental state. Indeed, we can consider that as something that does not facilitate the investigation of mental function, whose time is much faster. Some tasks are very difficult to maintain for 45 minutes.

When you deal with aspects of synaptic activity that you like to refer to mental processes that are faster than the ones that you want to investigate, you can rely on markers that primarily are related to changes in blood flow supply, whose activity is much faster.

Example: H_2^{15}O half-life is 2 minutes, so in 10-15 minutes the activity of oxygen is disappeared, so you can monitor changes in blood flow that occur in few seconds. In some cases, oxygenated blood reaches a peak in the brain, that is compatible with several neural functions. Changes in blood flow supply are tightly coupled in changes in glucose and oxygen consumption (synaptic activities), so it can be used to represent synaptic activity, but you use tracers with different metabolic fate time.

When I'm dealing with an asymmetric distribution of glucose consumption and I'd like to compare these, you need to have absolute values for each of the voxels. Once you know the amount of dose that you have injected and the body mass of the participant, then you are able to reconstruct the curve, interpolating the values. Therefore, you are able to express Deoxyglucose as mg/100g tissue/minute (generally the range of value is 10-15 mg) or ml/100g tissue/minute (generally 50-70 ml).

Once you have the absolute value, we have the parametric or dynamic image, and at that point we can do any statistical operation with these images. When you take several volumes, you are trying to do an interpolation, so that you can have images to overlap. An important step is the spatial normalization, so that each we can move our brain into a standard reference state and have the possibility to compare different realizations (for example from different people). Each point is characterized by 3 coordinates. We use ATLAS (space of coordinates of reference), the first was the Talairach, obtained cutting the brain in slices and for many years it was the main reference for all brain studies.

In functional neuroimaging studies, cognitive subtraction refers to an aspect of experimental design involving the comparison of two conditions or brain states that are presumed to differ in only one discrete feature (the independent variable). Pure insertion asserts that there are no interactions among the cognitive components of a task

Donders' subtractive Methods

The main approach that we got derives from experimental psychology, that grew up from subtraction approaches.

Donders' work attempts to describe the processes going on in the mind, by analyzing cognitive activity into separate stages (many scientists assumed that responding to a stimulus occurred instantaneously!)

Donders was particularly interested in "timing the mind" and used a subtraction technique to time the different mental processes that the brain goes through when faced with different tasks.

Donders performed experiments using reaction time tasks in 1868:

1. a simple reaction time task - e.g. you are seated in front of a panel that contains a light bulb and a response button. When the light comes on, you must press the button. 190 ms

2. a discrimination reaction time task - e.g. you are seated in front of a panel with 5 light bulbs and one response button. When the target light goes on you must press the button - but not if the 4 other lights come on. 240 ms

These 50 ms of difference is the time that the mental function needs to discriminate the target.

3. a choice reaction time task - e.g. you are seated in front of 5 light bulbs, each with their own button. You must press the button corresponding to the appropriate light. 280 ms

Together with discrimination you have to select the correct motor action. The idea is that as soon as you add pieces, your reaction time is longer and you can build up experiments that allow you to add piece, that you can subtract from another condition.

Example: For visual stimulation, eyes open and white light or control condition and eyes closed. The only difference is the perception of the white light. You subtract statistically the PET images, and you obtain a group average response, and the mean difference image is something that tells you that perception of the white light relies on changes in activity in the occipital cortex.

Example: Chess as a case of study. Why chess?

- Chess problems are complex.
- The ability of the chess player is measurable.
- There are multiple studies in which the cognitive psychology of chess is studied.

In this study they decided to split the chess game in 4 different mental stages that we could compare:

- B/W discrimination.

Control baseline condition. The idea was to define specifically what the brain is doing during the tasks. (Visual perception, discrimination of colors, reading, recalling motor command, decision, ocular movement, thumb movement).

- Spatial discrimination (is the closest piece to the X white?).

All the things we said before but with an additional condition, spatial analysis of distance. (Spatial attention, static spatial analysis). Subtracting the PET images, we are able to see the different areas of the brain working in the new task. (Precuneus, occipital lobe, superior parietal lobule, medium temporal gyrus, superior premotor cortex).

- Rules (can the white queen catch the black pawn).

You have to discriminate the value of the pieces and recall the rules, so you must activate the memory. (Value discrimination, rules recall, dynamic spatial analysis). The areas of the brain working are different (left hippocamp, left temporal lobe, left superior temporal gyrus, left medium temporal gyrus, left fusiform gyrus, right post-central gyrus, cerebellum).

- Checkmate (can the white player win in one move).

(Recall of the game rules for all the pieces, consequence valuation for every move, generation of moves sequence). This was a 1994 study.

Example: Four different experimental conditions, in which order we should make the patient do the different tasks.

- Visual
- Spatial
- Rules (memory)
- Strategy

The first thing we would say is to make him do them in this exact order, but we have to take in consideration the increase of difficulty for the different tasks. How can we discriminate between the areas of the brain that are working trying to solve a more difficult task (involving emotional harassment) and the ones specifically working for the task itself. How to manage task effect? They also learn to do the tasks, so the tasks become easier and brain response is somehow reduced, or you can become tired and habituated. We can't present them in casual order because we would have to compare different patients that had the tasks presented in different order.

You teach the patients the tasks, until they reach maybe 90% accuracy, and then you can have your experiment putting them in the machine. In this way you avoid training effect.

Face perception

Example (How to build a PET imaging experiment)

Which are the variables that we have to take into account to determine our patients?

- Gender (sexual orientation)
- Ethnicity (cross-race effect)
- Age
- Education
- Health status
- Drug-free
- Psychometric characterization
- Possible selection bias (mostly students)
- Sampling methodology
- Calculating sample size

Cross-race effect: when we are asked to recognize the identity of people that we are not familiar with (Chinese, afro Americans), our performance is decreased in respect to recognizing Caucasians for example.

Experimental paradigm (How can we define the stimuli?)

The stimuli allow to study what is the GESTALT, psychology that focuses on how humans naturally perceive patterns and organize sensory information into meaningful wholes, rather than just as collections of separate parts.

One element is important; we have to discriminate low-level features of vision (the first kinds of information that our visual system extracts from the raw input that hits the retina). You want to have a stimulus that allows you to compare two stimuli with the same features (contrast, luminosity), so that the only difference is only the lack of a face.

We can have different regions of the brain involved with respect to the stimuli that is presented:

- Faces vs houses → Semantic
- Active vs passive → Attention
- Picture vs scrambled → Mid-level

Scrambled: a visual stimulus that has been disorganized meaning its low-level features (like color, brightness, or texture) are preserved, but its meaningful structure or recognizable shape is destroyed.

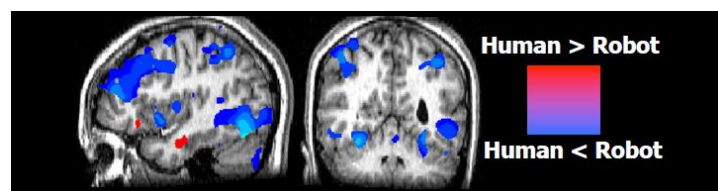
Everything that is in common is specific to face recognition (junction analysis), in this case the fusiform (one of the gyrus of the ventral surface of the temporal lobe) face area. One thing that you want to be aware of is that there may be something that biases the representation of faces, difference in contrast or luminosity, the brain catches low-level features of vision.

Pharmacokinetics is how the drug enters the body, the pharmacodynamics is different, what receptor the drug reaches or which neurotransmitter it stimulates. Serotonin receptors are primarily located in the pre-frontal cortex; dopamine is primarily present in the striate cortex and Parkinson disease is characterized by the reduction of dopamine levels (detectable with PET images).

Perception and sensation

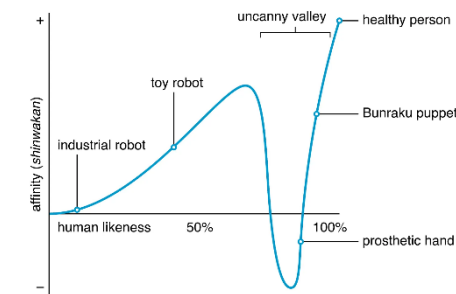
In the years we were able to see how visual perception affects the brain, how the occipital works.

It's important to study face perception not only in humans, but also in machines. An example is the project of Takanishi Laboratory.



People have been working hard trying to have precise details, but we struggle to say that this could be a face. Also, our brain reacts in different ways, there are two regions (one in the temporal lobe and one in the frontal cortex), associated with social cognition and interaction.

The Uncanny Valley is a phenomenon where something that looks almost human, but not quite perfectly human, evokes feelings of unease, discomfort, or eeriness in observers.

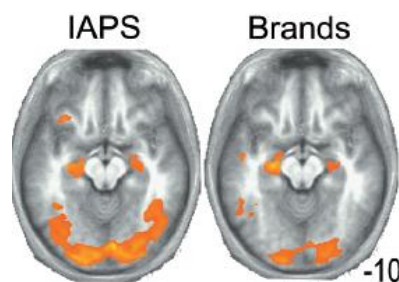


Source: Masahiro Mori, "The Uncanny Valley," *IEEE Robotics & Automation Magazine*, 19(2):98-100 (June 6, 2012).

Uncanny Valley phenomenon curve.

The International Affective Picture System (IAPS) is a database of pictures designed to provide a standardized set of pictures for studying emotion and attention that has been widely used in psychological research. They got images and asked how positive or negative these scores are.

Each picture was shown every 2 seconds (positive, negative and brand logos), what people was asked to do was to pay attention to the stimuli and then press the orange button (Oddball task). Looking at the brain's images, we see the visual perception regions activated, but also the amygdala, labelled as emotional sentinel. Even when you recognize something that you like, like a logo, your brain activates a pleasant score as a reward.



Perception should get piece of information just to reproduce and to survive, not to get a replica of the world. For instance, the frog has a great ability to discriminate between different tonalities of few colors (blue and green), but is totally blind to others, because they live in pods. Sensory modalities evolved to guide animal behavioral choices, necessary to reproduce/survive.

Our brain has to alter the kind of perception (change the colors or the proximity), just like top-down modulation, that tells the brain to focus and adapt to the perception.

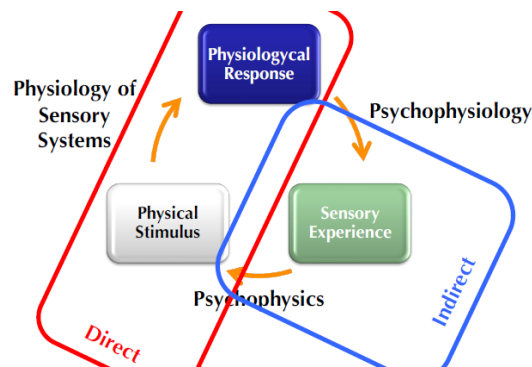
- Top-down perception is when your brain uses what you already know — your experiences, expectations, and context — to understand what you sense.
- Bottom-up perception is when understanding starts with the raw information from your senses, and your brain builds meaning from that data.

The difference between the way we interpret, and elaborate information is what we refer to as perception.

The way we see and represent colors is individual, but the wavelength of the color is the same for everybody. This is telling us that we can share some of this subjective representation but it remains different for each of us.

We have more than five senses, but all of them share 3 common stats, there is a physical stimulus that arrives, a second event transforms this stimulus into a nerve impulse (action potential), and

then there is a response in the form of a sensation or conscious experience of sensation. These are 3 steps that are acknowledged.



Early findings in psychophysics and sensory physiology exposed a weakness in the empiricist argument: the mind is not a tabula rasa. Our perceptions differ qualitatively from physical properties — colours, tones, and smells are mental creations constructed by the brain from sensory experience.

Nonetheless, our perceptions correspond to the physical properties of stimuli. Knowledge, therefore, is based not simply on sensory experience but also on the brain's inherent properties that organize that experience — echoing Kant's idea of a priori knowledge.

Since the first studies each stimulus could be characterized with 4 features:

- **Modality:** general class of stimulus, determined by the type of energy transmitted by the stimulus (light, heat), and it is defined by the receptors that capture it.
- **Location:** the set of sensory receptors that are active, and that are actually telling me that specific energy derives from a certain source.
- **Intensity:** the response amplitude, primarily related to the stimulus energy and the transformation of this energy into nervous impulses. The higher energy the higher the nervous transduction is.
- **Timing:** when stimulation starts and stops.

In each sensory system the initial contact with the external world occurs to specialize structures that we call receptors. The sensory receptors are the first cells of the sensory process that transform the stimulus energy into an electrical signal, this is called stimulus transduction. Each receptor specifically responds to one form of energy.

All receptors have different receiving fields, smaller or broader, determining the ability (together with the receptor density) of discriminating two stimuli and separate in space and time. The density of receptors defines the resolution across different body parts (fingertips or knee).

This type of labelled code line is not something that is equal across all different sensory modalities, there is also that all signal from sensors has to reach the brain, and the detail of this action potential tell the brain which type of energy was received, the timing and the location. All sensor modalities have central processing, from the thalamus to the cortex. All the information we gather from the thalami is called primary sensory areas. Only olfaction does not pass through the thalamus before reaching the cortex.

We are talking about sensation, that comes from the periphery, able to create a population code, a sequence of connection for example along the skin and the spinal cord, or the retina for the

eyes, a hierarchical organization and extraction of information to convey to the brain and then process the information.

Several factors operate to shape and sometimes distort perception. These factors can reside in the perceiver, in the object or target being perceived, or in the context of the situation in which the perception is made. This happens for example at the level of the retina for some type of illusions.

We see things as we are, and this is what we really call perception. It's not just the process of receiving the information but also making sense of this piece of information, this is how experience works. Perception can be influenced by several things, like physiological state (food when I'm hungry) or the context (crosta o crumble di pane). It's plenty of things that can alter our perception of things (just like blue risotto).

Organization is the process of placing selected stimuli in a framework, providing categorical recognition.

The f-MRI

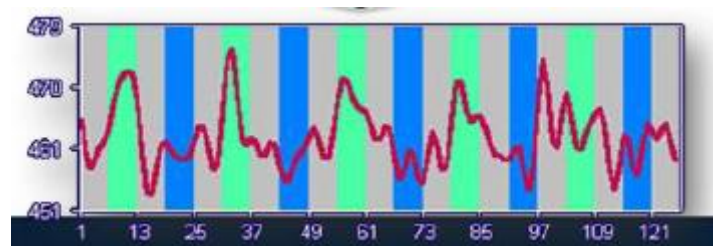
MRI studies brain anatomy, f-MRI studies the functioning. MRI is primarily using one image with high resolution, for f-MRI we acquire different images in time (every 2 sec for 5 minutes) with low resolution. Comparing it with PET imaging we are adding the temporal component.

Nuclear magnetic resonance is based the electromagnetic stimulation that deviates, flips the proton, and once the stimulus is over you let it realign with the magnetic field with a specific relaxation time, that influences different aspects.

We are dealing with endogenous contrast, the signal that we are measuring is the BOLD signal (**Blood Oxygenation Level Dependent signal**). When you have stimulation, you have a higher consumption of glucose and higher blood flow, but the flux of oxygen is not proportional to the amount of oxygen that the brain receives (luxury perfusion, it arrives more blood than needed).

Oxygen and carbon dioxide are bonded to one protein in the red blood cells, oxyhemoglobin, that has different magnetic properties, is slightly diamagnetic (attractive) while deoxyhemoglobin is paramagnetic. So, you have an indigenous tracer that has two different behaviors, and the T2 (relative to the interaction with hemoglobin) is reducing the signal when there is a higher concentration of deoxyhemoglobin (slightly attractive), so that in resting condition you have an unbalance between the arterial side and the venous side.

When I have a stimulus that activates the brain, I have a lot of oxygenated blood (more blood flow than oxygen consumption), and while in the arterial side there are no changes in the venous side something happens. This is because some oxygenated blood gets into the venous side, and becomes more diamagnetic, increasing the signal also in this side.



Cleaned signal in which we can see the stimulus.